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| Name of the Student | G. Naga Mani Kanta |
| Reg. No | 192425310. |
| GitHub Link | |

28/8/25

**SIMATS ENGINEERING**

**CO2 - ASSESSMENT TOOL 3**

# Class Test

**COURSE NAME:** **Deep Learning** **COURSE CODE:** **MLA0405**
**Faculty : Dr. Arul Raj A.M** **Slot :**

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**COURSE OUTCOME COVERED**

**CO 2:** Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

**CO Weightage : 15%**
**Assessment Tool Weightage: 20%**
**Duration: 90 minutes**
**Marks: 25**

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**Code of Conduct:**

I G. Naga Mani Kanta (192425310) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

G. ManiKanta G. Naga Manikanta

**Signature of the student:** **Name of the student:**

| Criteria | Conceptual Understanding (2.5) | Linkages (2.5) | Organization (2) | Integration of Literature (2) | Clarity & Presentation (1) | Total (10) |
|---|---|---|---|---|---|---|
| Weightage | 25 % | 25% | 20% | 20 % | 10 % | 100% |
| Q1 | | | | | | |
| Q2 | | | | | | |
| Q3 | | | | | | |
| Q4 | | | | | | |
| Q5 | | | | | | |
| | | | | | | |
| Total | | | | | | |

**Class Test**

**CO2-AT3**
**QUESTIONS: 05**

**Total Marks:25**

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of **Representation Learning**. Discuss how the **width** and **depth** of a neural network influence its learning capability, computational complexity and model performance.
3. Compare **ReLU, Leaky ReLU (LReLU)** and **ELU (Exponential Linear Unit)** activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of **unsupervised learning** in neural networks. Describe the working principle of **Restricted Boltzmann Machines (RBMs)** and **Autoencoders** and compare their objectives and applications.
5. Explain the concept of Autoencoder and its uses in Deep learning .

**RUBRICS FOR CALCULATION:**

| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
|---|---|---|---|---|---|
| Concept Identification | 25% | Identifies all key concepts from literature accurately and comprehensively | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts |
| Linkages | 25% | Establishes clear, meaningful, and accurate relationships between concepts | Shows correct relationships with minor gaps | Relationships are weak, unclear, or partially incorrect | No meaningful relationships established |
| Organization | 20% | Concepts are well-structured hierarchically with logical flow and grouping | Mostly organized with minor structural issues | Some organization but lacks clear hierarchy | No logical organization or structure |
| Integration of Literature | 20% | Integrates multiple studies effectively to show connections and comparisons | Shows some integration of literature | Limited integration; mostly isolated concepts | No integration of literature evidence |

G. Naga Mani kanta
192425310.

(1) Machine Learning :- Machine learning ~~is~~ a subset of AI which learns from data & make decisions or predictions without explicitly being Programmed by the Programmer.

Deep Learning :- Deep learning is a subset of Machine learning where computer learns from complex neural tasks using hidden layers automatically.

Real world examples :-

ML :- ① Fraud-detection in Credit card management.

② fake money transfers through UPI.

Deep Learning :- ① Face-recognition in Phone locks.

② used in Self driving cars to detect objects around them.

| Method | ML | Deep learning. |
|---|---|---|
| Architecture | It is simple. | It is Complex with many layers. |
| Data-requirements | Small data Samples might be okay. | It require large Data Samples. |
| Feature extraction | It uses algorithms like KNN, random forest etc.. | It uses advanced neural networks. |
| Applications | Spam emails, Fraud Payments and Bank frauds etc.... | Face-recognization, speech-recognization etc... |

(2) Representation Learning:- Representational learning is a process where neural networks automatically learns useful features or representations from raw data.

Instead of manually selecting them, the networks learns themselves.

width:- The number of neurons in a layer is called width.

A wider network can learn more Patterns at the same layer.

Advantages:-

(i) Can represent more complex Patterns.

(ii) May improve model Performance.

Disadvantages:-

- Requires more memory.
- Increase computation.

Depth:- The number of layers in a neural network is called Depth.

Different layers can learn hierarchical representations.

③ ReLU vs Leaky ReLU vs ELU.

① ReLU:- It is Rectified Linear unit.

$$f(x) = \max(0, x)$$

Advantages:-

(i) It is simple.

(ii) Computionally fast.

(iii) Helps reduce the vanishing-gradient Problem for +ve values.

Disadvantages:- Dying ReLU Problem.

② Leaky ReLU:-

It is designed to solve the dying ReLU Problem.

$$f(x) = \begin{cases} x, & x > 0 \\ \alpha x, & x \leq 0 \end{cases}$$

usually, $\alpha$ is a small value such as 0.01.

Advantages:-

(i) Reduces dying ReLU Problem.

(ii) Allows a small gradient for -ve values.

Disadvantages:- • Requires choosing the value of $\alpha$.

③ ELU:- It stands for Exponential Linear unit.

$$f(x) = \begin{cases} x & x > 0 \\ a(e^x - 1) & x \leq 0. \end{cases}$$

(+ve ELU behaves like ReLU)

(-ve, it smoothly goes to $-\alpha$.)

Advantages:- (i) Has smooth -ve values.

Disadvantages:- (i) More computationally expensive than ReLU.

(ii) Uses an exponential calculation.

(4) Unsupervised Learning:-

In unsupervised learning, the models learn from data without labeled outputs.

Restricted Boltzmann Machines (RBMS):-

It is a neural network used for unsupervised learning & representational learning.

It has two layers:-

① Visible layer.

② Hidden layer.

Working:- (i) Input data is given to the visible layer.

(ii) Information is passed into the hidden layer.

(iii) Hidden neuron learn important patterns.

(iv) The model tries to reconstruct the original data

(v) Difference between original & reconstructed data help train the data.

Autoencoder:- It is a neural network that learns to represt input data in a smaller or meaningful form to reconstruct the original data.

(5) **Autoencoder:-** An autoencoder is a type of neural network that learns to compress input data into a smaller representation and then reconstruct it.

(i) **Encoder:-** Takes the input & convert it to a small representation.

(ii) **Latent Representation:**

This is the compressed representation of the input.

Input → Encoder → Latent Representation.

(iii) **Decoder:-**

The decoder takes the compressed representation and tries to reconstruct the original input.

uses of **Autoencoder:-**

(i) Dimensionality reduction.

(ii) Data compression.

(iii) Noise ~~removal~~ removal.

(iv) Anamoly Detection.

(v) feature learning.